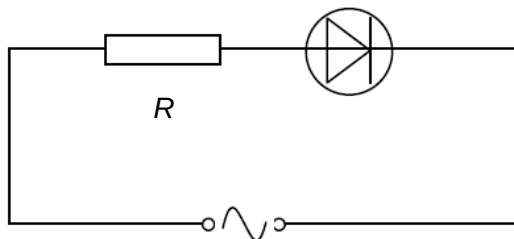


25

A sinusoidal current of peak value I_0 is passed through the circuit shown.



What is the mean rate of heat dissipated in terms of R ?

A $1.41 I_0^2 R$

B $I_0^2 R$

C $0.5 I_0^2 R$

D $0.25 I_0^2 R$

Ans: D

$$P = I_{\text{rms}}^2 R$$

With a diode in the circuit, half-wave rectified.

Normally sinusoidal current, $I_{\text{rms}} = I_0/\sqrt{2} \rightarrow I_{\text{rms}}^2 = I_0^2/2$

But for half-wave rectified, $P = \frac{1}{2} (I_0^2/2) R = 0.25 I_0^2 R$

26

The ratio of turns of the primary coil to the secondary coil of a practical transformer is 10:1